

LONGYAN, China

WMO: 589270

Lat: 25.100N

Lon: 117.017E

Elev: 1119

StdP: 14.11

Time Zone: 8.00 (E08)

Period: 82-93

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	36.5	39.8	21.0	16.4	51.0	25.7	20.5	48.6	13.9	54.5	12.1	53.5	3.1	0	0.297

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	94.2	75.4	92.5	75.4	90.9	75.2	77.8	87.8	77.2	87.0	76.6	86.0	6.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.5	139.2	81.0	74.8	136.0	80.5	74.4	133.9	80.3	42.2	88.1	41.6	87.2	41.1	86.4	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	10.9	9.6	DB	31.9	98.3	3.1	1.6	29.6	99.4	27.8	100.4	26.0	101.3	23.7	102.5	(4)
(5)			WB	29.1	80.7	2.7	1.8	27.1	81.9	25.5	83.0	23.9	84.0	22.0	85.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	69.3	54.4	56.0	61.1	68.3	74.9	79.4	82.9	82.3	78.7	72.8	64.0	55.8	(6)
(7)		DBStd	11.60	6.44	6.73	7.70	6.38	4.91	3.92	2.79	2.65	3.99	4.91	5.77	6.43	(7)
(8)		HDD50	77	32	15	7	0	0	0	0	0	0	0	1	22	(8)
(9)		HDD65	1180	330	259	170	39	3	0	0	0	0	4	86	289	(9)
(10)		CDD50	7114	168	184	349	548	771	883	1019	1000	861	706	422	203	(10)
(11)		CDD65	2743	2	8	47	137	310	433	554	535	411	245	56	5	(11)
(12)		CDH74	23147	4	28	179	698	1999	3788	5977	5377	3307	1505	267	18	(12)
(13)		CDH80	8416	0	1	16	145	588	1270	2602	2211	1134	416	33	0	(13)
(14)	Wind	WSAvg	3.6	3.5	3.4	3.5	3.3	3.5	3.9	4.1	3.5	3.5	4.1	3.8	3.2	(14)
(15)	Precipitation	PrecAvg	67.4	2.2	3.7	7.0	7.9	11.5	11.9	5.1	7.1	5.3	2.3	1.8	1.3	(15)
(16)		PrecMax	98.3	8.8	13.8	22.5	19.3	22.0	21.9	13.5	15.6	16.7	7.1	5.0	4.6	(16)
(17)		PrecMin	48.1	0.0	0.0	0.4	2.4	2.2	3.6	1.6	1.1	1.1	0.0	0.0	0.0	(17)
(18)		PrecStd	11.6	2.4	3.0	4.7	4.1	4.7	4.9	2.5	3.7	3.5	1.9	1.5	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.5	77.4	82.2	87.7	91.2	94.3	97.6	95.4	93.5	91.2	83.7	76.5	(19)
(20)			MCWB	61.5	62.2	67.8	70.7	73.7	75.7	75.6	74.9	76.1	73.5	68.7	62.1	(20)
(21)		2%	DB	71.1	74.1	79.1	84.1	88.2	91.5	94.7	93.6	90.7	87.6	80.2	72.8	(21)
(22)			MCWB	59.1	61.9	67.1	70.2	73.1	75.3	75.6	75.1	75.2	71.8	65.9	59.8	(22)
(23)		5%	DB	68.2	70.7	75.8	81.0	86.0	89.3	92.8	91.9	88.6	84.5	77.0	69.7	(23)
(24)			MCWB	57.9	60.2	66.3	69.3	72.6	75.3	75.6	75.1	74.3	70.1	64.7	57.7	(24)
(25)		10%	DB	65.0	67.1	72.4	78.0	83.4	87.0	91.0	90.2	86.5	81.8	73.8	66.7	(25)
(26)			MCWB	56.3	59.5	65.0	68.6	72.1	75.0	75.7	75.3	73.4	68.2	62.4	56.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.1	66.0	70.1	73.8	76.4	78.2	78.6	78.9	77.8	75.0	70.8	63.6	(27)
(28)			MCD B	69.6	73.1	78.6	83.3	84.3	86.8	89.9	89.1	88.4	87.6	80.1	72.8	(28)
(29)		2%	WB	61.5	63.9	68.5	72.2	75.2	77.3	77.5	77.8	76.9	73.6	68.5	61.6	(29)
(30)			MCD B	68.5	70.3	75.5	80.3	83.7	85.8	88.3	87.7	86.7	84.5	76.2	70.2	(30)
(31)		5%	WB	59.5	62.5	67.4	71.2	74.4	76.8	77.0	77.0	76.2	72.4	67.0	59.7	(31)
(32)			MCD B	66.1	67.8	73.8	78.6	82.2	85.0	87.6	86.8	85.0	81.1	73.5	67.0	(32)
(33)		10%	WB	57.5	60.8	66.0	70.2	73.6	76.2	76.4	76.4	75.4	71.0	65.2	57.9	(33)
(34)			MCD B	63.4	66.1	71.9	76.7	80.7	83.9	86.4	85.4	83.2	78.3	71.9	64.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	14.8	12.1	11.2	11.7	12.2	12.1	15.3	14.9	13.5	15.4	16.4	17.7	(35)
(36)			MCDBR	20.3	19.6	17.3	16.6	17.1	16.1	19.2	18.4	17.4	19.2	20.2	21.6	(36)
(37)			MCWBR	9.8	9.2	7.2	6.0	5.5	4.1	4.6	4.4	4.7	6.1	7.9	9.6	(37)
(38)		5% WB	MCD BR	15.8	14.3	13.9	12.9	13.5	12.7	15.9	15.3	14.7	16.5	16.4	17.0	(38)
(39)			MCWBR	8.2	7.7	6.6	5.5	5.1	4.3	4.5	4.7	4.5	5.9	7.1	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.568	0.638	0.753	0.744	0.622	0.576	0.550	0.616	0.630	0.617	0.544	0.527	(40)	
(41)		taud	1.760	1.658	1.484	1.546	1.823	1.944	2.014	1.824	1.771	1.744	1.875	1.870	(41)	
(42)		Ebn at Noon	213	209	193	201	227	236	242	226	218	211	218	218	(42)	
(43)		Edh at Noon	64	75	94	90	68	60	56	68	69	68	56	55	(43)	
(44)	All-Sky Solar Radiation	RadAvg	866	904	988	1117	1226	1317	1686	1536	1398	1250	993	921	(44)	
(45)		RadStd	146	171	150	179	196	160	151	146	126	187	127	153	45	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		+0.58	N/A	N/A	+0.90	N/A	N/A	N/A	N/A	+198	+160

Nomenclature: See separate page